OIL SPILLS & SHIPPING SAFETY

Institutional Analysis of the Prestige (2002)
Comparing Arrangements to Address the Social Dilemma

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Source: Yahoo News / AFP

The Prestige (2002) and Institutional Response

77,000 t
heavy fuel oil spilled

2,900 km
coastline contaminated

~€4 B
total estimated damage

€171.5 M
maximum compensation paid



Source: Yahoo News / AFP

The institutional response

The EU accelerated the single-hull phase-out (MARPOL Annex I) and the 2003 Supplementary Fund Protocol raised the compensation ceiling from SDR 203 M to SDR 750 M. Yet the question is whether these reforms addressed the underlying incentive failure.

EXTERNALITIES

Four Types and the Liability Gap

Negative Production

Safety underinvestment is a cost-reducing input decision. Spill costs fall on third parties outside the market transaction.

Transboundary

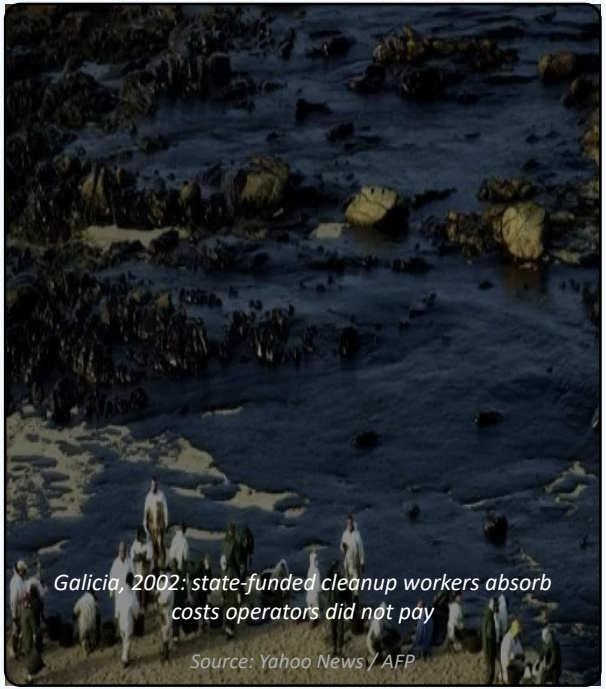
A Bahamian vessel, a Russian charterer, damage in Spain and France: no single jurisdiction holds authority over the full harm.

Intergenerational

Degraded fisheries and benthic ecosystems persist for decades. Future generations bear costs with no representation in current decisions.

Pecuniary

Lower freight rates diffuse benefits widely across consumers. This weakens political pressure for reform — those who gain do not identify themselves as beneficiaries.



Galicia, 2002: state-funded cleanup workers absorb costs operators did not pay

Source: Yahoo News / AFP

The liability gap: €171.5 M paid against approximately €4 B in damage. Operators covered approximately 4% of verified losses. The gap is not incidental; it is structurally produced by GS5.

SOCIAL DILEMMA

A Prisoner's Dilemma Among Tanker Operators

The Operator's Choice

If one operator invests in safety, others do not:

The investing firm loses freight contracts on price. Responsible behaviour is penalised by market selection.

If all operators reduce safety expenditure:

Industry-wide risk rises. The outcome is collectively worse, yet individually rational.

Dominant strategy:

Underinvest regardless of competitor behaviour. The defection equilibrium is stable.

Why Self-Governance Cannot Resolve This

A8: Zero resource dependence

Operators vacate the affected area after each voyage. There is no long-term stake in the condition of the ocean they pollute.

GS5: Liability cap

Capped financial exposure means underinvestment remains profitable even after a spill occurs.

A1: Anonymous actors

Hundreds of shell-company operators preclude reputational enforcement or mutual monitoring.

SES FRAMEWORK · TWO CORE VARIABLES

A8 + GS5: The Closed Loop of Failure

A8 Resource Dependence

GENERATES THE DILEMMA

Shipping operators have no long-term dependence on the specific ocean they may contaminate; they move on after each voyage.

In Ostrom's successful cases such as Swiss alpine pastures and Maine lobster fisheries, users remain in the resource system and therefore internalise future consequences.

Absence of dependence eliminates the intrinsic motivation for conservation.

GS5 Operational-Choice Rules

LOCKS IN THE DILEMMA

Liability caps and flags of convenience sever the financial connection between industry decisions and social costs.

In the Prestige case: the maximum compensation payable was €171.5 M against approximately €4 B in damage. The rules did not fail; they performed precisely as designed.

Removing the cap would make the externalising strategy financially non-viable. GS5 is an active cost-transfer mechanism.

Remove A8's effect: intrinsic motivation returns. Remove GS5: financial deterrence returns. Remove both: safety investment becomes rational.

ARRANGEMENT 1

IOPC Supplementary Fund Protocol (2003)

Mechanism

- A third-tier international fund, layered above the existing Civil Liability Convention (CLC) and IOPC Fund.
- Financed by oil importers receiving above 150,000 tonnes per year in participating states, not by vessel operators.
- Raised the total compensation ceiling from SDR 203 M (approximately €171.5 M) to SDR 750 M (approximately €871 M at adoption).
- Adopted following the Erika (1999) and Prestige (2002) spills; entered into force May 2005.

Key Numbers

€171.5 M
pre-2003 ceiling

€871 M
post-2003 ceiling

×5.1
increase in coverage

~4%
Prestige coverage ratio

SES diagnosis: The fund directly targets GS5 by raising the financial ceiling on post-incident compensation. It does not alter operator incentives ex ante (A8 remains intact) and is financed by oil importers rather than by those making the safety decision.

ARRANGEMENT 2

Market Consolidation and Vertical Integration

The Argument

- With hundreds of anonymous shell-company operators, reputational enforcement is structurally impossible. Consolidation creates identifiable actors with sustained reputational stakes.
- Major oil companies (Shell, BP, Total) operate vetting programmes (SIRE) under which only inspected vessels are chartered. This constitutes de facto consolidation at the charterer level, raising the safety floor above the IMO minimum.
- Vertical integration, where cargo owners operate their own tankers, eliminates information asymmetry in the procurement market entirely.
- A concentrated operator landscape increases the cost of a spill event beyond the formal CLC cap through investor responses, charter market exclusion, and insurance withdrawal.

Mechanism

Reduces A1

Fewer anonymous actors; mutual visibility increases.

Closes A8

Firm reputation becomes contingent on the future condition of the ocean.

Privatises GS5

Reputational costs function as an informal, uncapped liability.

SES diagnosis: Consolidation addresses A8 and partially GS5 through reputational stakes, creating a future stake in ocean quality. The arrangement is not universally enforceable and does not provide coverage for residual small or state-owned operators outside voluntary vetting programmes.

COMPARISON

Two Evaluation Criteria: Compensation Coverage · Prevention Incentive

	Arrangement 1 IOPC Supplementary Fund	Arrangement 2 Market Consolidation
Criterion 1 Compensation Coverage (GS5)	★★★★☆ The ceiling rises approximately fivefold to €871 M. The liability gap identified under GS5 is directly reduced, though coverage remains below actual damage in a Prestige-scale event.	★★☆☆☆ No formal compensation mechanism is created. Reputational pressure may incentivise settlements, but a small operator can still declare insolvency. The gap persists for residual actors.
Criterion 2 Prevention Incentive (A8 + GS5)	★★☆☆☆ The fund is financed by oil importers, not by operators. Spill consequences for the operator remain capped at the CLC level. The ex ante safety calculus is unchanged; A8 is fully intact.	★★★★☆ Reputational stakes internalise future ocean risk into firm value. Operators dependent on major charterers invest in safety to retain market access. Both A8 and informal GS5 are partially addressed.

Key finding: the two arrangements are complements. Arrangement 1 addresses the ex post compensation gap (GS5). Arrangement 2 addresses ex ante operator incentives (A8). Neither is sufficient in isolation.

CONCLUSION

The liability gap is an institutional design choice, not an incidental outcome.

01

Social Dilemma

Operators face a stable defection equilibrium. A8 eliminates intrinsic motivation; GS5 eliminates financial deterrence. External intervention is the only viable corrective path.

02

Arrangement 1

The Supplementary Fund (2003) raised the compensation ceiling fivefold. It addresses GS5 ex post but leaves A8 and the operator's safety calculus unchanged.

03

Arrangement 2

Market consolidation, evidenced by the SIRE vetting system, creates reputational stakes that partially internalise A8 and constitute an informal substitute for uncapped liability.

Together they are complements: the fund covers the aftermath; consolidation changes the incentive to cause harm in the first place.

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